

Lecture #10, Oct 16, 2025

Boundary condition

1) Dirichlet boundary condition

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}, \quad x \in [0, 1], \quad \nu > 0$$

$$u(0, t) = \alpha$$

$$u(1, t) = \beta$$

Solution at boundary is given

Special case

$$\frac{\partial u}{\partial t} = 0 \quad \text{no time dependence}$$

B.c. are homogeneous, $\alpha = 0 = \beta$

$$0 = \nu \frac{\partial^2 u}{\partial x^2}$$

$$\frac{\partial^2 u}{\partial x^2} = 0$$

Fourier mode. $e^{ikx} = \cos kx + i \sin kx$

Since $\cos kx \neq 0$ at $x=0$

expect only $\sin kx$ to form solution

Better: $\frac{\partial^2 u}{\partial x^2} + \lambda u = 0 \quad (\text{ODE})$

2) Neumann boundary condition.

normal component of derivative is given

$$\text{at } x=0, \quad \nabla u \cdot \hat{n} = \frac{\partial u}{\partial x}$$

So

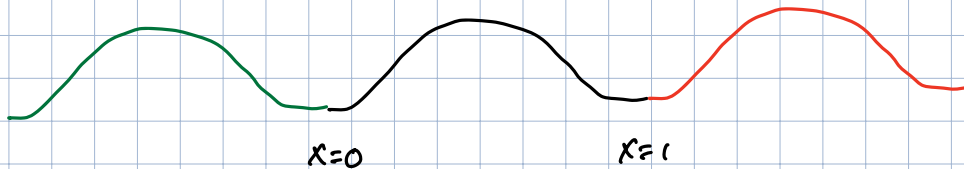
$$\left\{ \begin{array}{ll} \frac{\partial u}{\partial x} = \alpha & \text{at } x=0 \\ \frac{\partial u}{\partial x} = \beta & \text{at } x=1 \end{array} \right.$$

Since $\frac{\partial \cos kn}{\partial x} = -k \sin kn$ and $\frac{\partial \sin kn}{\partial x} = k \cos kn$

$\cos kn$ are expected Fourier modes

3) Periodic Condition

Solution $u(x, t)$ is periodic in x .



$$u(x=0, t) = u(x=1, t)$$

4) Robin Condition : mix Dirichlet and Neumann

$$\alpha u + \beta \frac{\partial u}{\partial x} = \gamma \quad \text{at } x=0$$

Goal: - Convert a PDE to a system of ODE given the boundary conditions (semi discrete form)

or Algebraic system if fully discretized in space/time

- Recognize the governing matrix.

$$\text{let } U(t) = [u(x_0, t), u(x_1, t), \dots, u(x_N, t)]^T$$

$$\text{for } x_j = j \Delta x, \quad j = 0 \dots N$$

Any linear PDE $\Rightarrow \frac{dU}{dt} = AU$

Let $U^n = [u(x_0, t_n), u(x_1, t_n), \dots, u(x_N, t_n)]^T$

Then PDE $\Rightarrow U^{n+1} = AU^n + F$
 $\uparrow$ boundary condition or other terms

Ex $\frac{\partial u}{\partial t} = v \frac{\partial^2 u}{\partial x^2} \quad x \in [0, 1]$

$\frac{\partial u}{\partial x} = f(t) \quad \text{at } x=0$

$\frac{\partial u}{\partial x} = g(t) \quad \text{at } x=1$

$x_j = j \Delta x \quad j=0 \dots N, \quad \Delta x = \frac{1}{N} = \frac{x_{\max} - x_{\min}}{N}$

Ghost point method (Leveque ch 9, 10)

At $x=0, j=0, \quad \frac{\partial u}{\partial x} \Big|_{j=0} = \frac{u_1 - u_{-1}}{2\Delta x} = f^n \dots \textcircled{0}$



At $x=1, j=N, \quad \frac{\partial u}{\partial x} \Big|_{j=N} = \frac{u_{N+1} - u_{N-1}}{2\Delta x} = g^n$

Here u_{-1} and u_{N+1} are called ghost point values.

Has $N+1$ grid points or degrees of freedom (dof)

two Neumann condition increases dof to $N+3$

need two more equations. Replace B.C. with PDE.

$$\left\{ \begin{array}{l} \text{At } j=0, \quad \frac{u_j^{n+1} - u_j^n}{\Delta t} = v \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2} \\ \frac{u_0^{n+1} - u_0^n}{\Delta t} = v \frac{u_1^n - 2u_0^n - u_{-1}^n}{\Delta x^2} \end{array} \right.$$

$$\text{At } j=0, \quad u_{-1}^n = u_1^n - 2\Delta x f^n \quad \dots \quad (1) \quad (\text{using } (0))$$

$$\text{PDE} \Rightarrow u_0^{n+1} = u_0^n + \frac{v\Delta t}{\Delta x^2} (u_1^n - 2u_0^n - u_{-1}^n) \quad \dots \quad (2)$$

Using (1) in (2)

$$u_0^{n+1} = u_0^n + D(u_1^n - 2u_0^n + u_1^n - 2\Delta x f^n)$$

$$\boxed{u_0^{n+1} = u_0^n + D(2u_1^n - 2u_0^n) - 2\Delta x D f^n}$$

$$\rightarrow = (1-2D)u_0^n + 2Du_1^n - 2\Delta x D f^n$$

Internal points: $j=1 \dots N-1$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = v \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}$$

$$u_j^{n+1} = u_j^n + \underbrace{\frac{v\Delta t}{\Delta x^2}}_D (u_{j+1}^n - 2u_j^n + u_{j-1}^n)$$

$$u_j^{n+1} = D u_{j-1}^n + (1-2D) u_j^n + D u_{j+1}^n$$

Similarly at $j=N$,

$$u_{N+1} = u_{N-1} + 2\Delta x g^n$$

$$\begin{aligned} u_N^{n+1} &= u_N^n + \frac{v\Delta t}{\Delta x^2} (u_{N+1}^n - 2u_N^n + u_{N-1}^n) \\ &= (1-2D)u_N^n + 2D u_{N-1}^n + 2D\Delta x g^n \end{aligned}$$

Final system

$$\begin{bmatrix} u_0^{n+1} \\ u_1^{n+1} \\ \vdots \\ u_N^{n+1} \end{bmatrix} = \begin{bmatrix} 1-2D & 2D & & & \\ 2D & 1-2D & 2D & & \\ & \ddots & \ddots & \ddots & \\ & & & 2D & 1-2D \end{bmatrix} \begin{bmatrix} u_0^n \\ u_1^n \\ \vdots \\ u_N^n \end{bmatrix} + \begin{bmatrix} -2\Delta x D S^n \\ 0 \\ 0 \\ \vdots \\ 2\Delta x D g^n \end{bmatrix}$$

$$\Rightarrow U^{n+1} = A U^n + F \quad \text{for non homogeneous B.C.}$$

$$\Rightarrow U^{n+1} = A U^n \quad \text{for homogeneous B.C.}$$

Properties-

1) A is called amplification matrix.

$$A = \frac{u^{n+1}}{u^n} \text{ is like } G$$

also called finite difference matrix.

also called propagation matrix

→ Data-driven or machine learning technique

$$2) \text{ We see that } U^{n+1} = A^n U^0 \approx \lambda_t^n U^0$$

Exo Derive the amplification matrix A for heat eq.
using $u(0)=0, u(1)=0$

Neumann B.C without using ghost point.

two options:

forward / backward schem.

$$\text{At } x=0, j=0, \quad \frac{\partial u}{\partial x} = \frac{u_1^n - u_0^n}{\Delta x} = \alpha \Rightarrow -u_0^n + u_1^n = \Delta x \alpha$$

$$\text{At } x=1, j=N, \quad \frac{\partial u}{\partial x} = \frac{u_N^n - u_{N-1}^n}{\Delta x} = \beta$$

$$A = \begin{bmatrix} -1 & 1 & & & & \\ 2D & 1-2D & 2D & & & \\ & \ddots & \ddots & \ddots & \ddots & \\ & & & & & -1 & 1 \end{bmatrix}$$

$$U^{n+1} = AU^n + F, \quad F = [\Delta x \alpha, 0, \dots, \Delta x \beta]^T$$

Problem spatial accuracy is $O(\Delta x)$ overall
simplest. instead of $O(\Delta x^2)$

Remark We could use one sided approximation

for $\frac{\partial u}{\partial x}$ at $x=0$ and $x=1$

(best approach)

(forward)

(backward)

Remark: Leveque 2.12 shows boundary condition
for steady state case
Lecture note follows this section.

Leveque chap 9 Eq 9.9 shows FD matrix
for equati.

See Eq 9.11 Regarding semidiscrete system
Eq 9.16 full discrete case

Read def 9.1, Thm 9.2